Cosmology

Exercises Sheet 3

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Exercise 1

a)

The energy density at the time of nucleosynthesis can be find assuming that the Universe is radiation dominated, and that the radiation density energy is given only by photon (relativistic particles). Hence for relativistic baryons the density energy is [1]:

$$\rho c^2 = \frac{\pi^2}{30} \frac{k_B^4}{\hbar c^3} g T^4 \tag{1}$$

Indeed, at the time of nucleosynthesis all was relativistic. George Gamov knew that nucleosynthesis must have taken place at a temperature $T_{nuc} \approx 10^9 K$, but he didn't know about the three families of neutrinos and their interactions in 1948, so he didn't take it into account. Consequently, he took as degeneracy g only that relating to photons i.e. $g = 2$. The energy density at the time of nucleosynthesis is:

$$\rho_{nuc} c^2 = 3,775 \cdot 10^{-16} g T_{nuc}^4 = 7,55 \cdot 10^{20} \quad \frac{\mathrm{J}}{\mathrm{m}^3} \tag{2}$$

We can also write the mass density ρ_{nuc}, that is:

$$\rho_{nuc} \approx 8,4 \quad \frac{\mathrm{g}}{\mathrm{cm}^3} \tag{3}$$

Comparing this with the mass density at the centre of the sun, that is:

$$\rho_{\odot} \approx 150 \quad \frac{\mathrm{g}}{\mathrm{cm}^3} \tag{4}$$

we note that the two densities are not so different from each other. In the early universe the condition was similar at the condition at the centre of the sun. Knowing that inside the sun is where nuclear reactions occur, this leads to say that there was a period in the universe when it was possible to have a cosmological nuclear fusion condition and thus the creation of nuclei.

b)

The Hubble parameter at the time of nucleosynthesis is given by the first Friedman equation.

Since we are in radiation domination, we neglect matter, curvature k and the cosmological constant Λ. So the first Friedman equation is:

$$H_{nuc} = \sqrt{\frac{8\pi G}{3}\rho_{nuc}} = 2,17 \cdot 10^{-3} \quad \mathrm{s}^{-1} = 6,69 \cdot 10^{16} \mathrm{km/s/Mpc} \tag{5}$$

where the conversion from s^{-1} to km/s/Mpc is given by $H_0 = 2,18 \cdot 10^{-18}$ Hz $= 67$ km/s/Mpc. Therefore, we can see that $H_{nuc} \gg H_0$. This indicates that the expansion of the universe was faster during primordial nucleosynthesis than the current expansion of the universe. The expansion of the universe took place at a higher speed due to the higher energy density. In fact, the higher energy density of the universe at the time of nucleosynthesis compared to today has a direct impact on the expansion rate of the universe, as predicted by the Friedmann equations.

c)

To compute the time when nucleosynthesis take place, we consider that during radiation domination the scale factor $a(t) = (t/t_0)^{1/2}$ and the Hubble parameter is $H = \dot{a}/a$. This leads to

$$H = \frac{\dot{a}}{a} = \frac{d}{dt}\left(\frac{t}{t_0}\right)^{1/2}\left(\frac{t_0}{t}\right)^{1/2} = \frac{t}{2} \longrightarrow t_{nuc} = \frac{1}{2H_{nuc}} = 230 \quad s \tag{6}$$

d)

During radiation domination the scale factor is inversely proportional to temperature, so $T \propto 1/a \propto t^{-1/2}$. From here, we can compute the current temperature of the radiation:

$$T_0 = T_{nuc}\left(\frac{t_{nuc}}{t_0}\right)^{1/2} = 27K \tag{7}$$

where we have used the preset time $t_0 \approx 10^{10}$ yr. The observed temperature of CMB is $T_{CMB} =$ 2,73 K, The value of 10^{10} years we have used for the time of the universe is a rough estimate of the time elapsed from primordial nucleosynthesis to the present. This value may be different from the one used in current calculations, which is based on the most recent and accurate data on the age of the universe. Furthermore, the larger order of magnitude stems from the fact that temperature measurements are not precise. If we had used $T_{nuc} \approx 10^{10}$ would have come exactly 2.7 K.

However, the result obtained is not wrong and is the one obtained by Gamow in 1948. It is also possible to derive the nucleosynthesis temperature in another equivalent way. In fact, as written in Eq. (5), in radiation domination:

$$H = \sqrt{\frac{8\pi G}{3}\rho} = \sqrt{\frac{8\pi G}{3}\frac{\pi^2}{30}\frac{k_B^4}{\hbar c^3}g(T)T^4} \propto \sqrt{g(T)}T^2 \propto \frac{1}{t} \tag{8}$$

Considering the various values of constants, we obtain:

$$\frac{T}{1\mathrm{MeV}} \approx 1.5 g^{-\frac{1}{4}}\left(\frac{1\ \mathrm{s}}{t}\right) \tag{9}$$

We recall that $t_0 \approx 10^{10}$ yr are equivalent to $t_0 \approx 3.154 \cdot 10^{17}$ s, so:

$$T_0 \approx 2,25 \cdot 10^{9} \ \mathrm{MeV} = 26,2K \tag{10}$$

As demonstrated, the two results are practically identical.

e)

The assumptions of the temperature $T_{nuc} \approx 10^9 K$ and the age of the Universe $t_0 \approx 10$ Gyr are due to the knowledge of that epoch. Gamow and his PhD student Ralph Alpher assumed that the very early Universe was filled with a hot, dense gas of neutrons. When the space expanded the pressure fell and the neutrons started to decay into protons and electrons. Neutrons and protons could then fuse to deuterium nuclei. This was the start of the process of nucleosynthesis. After deuterium was created, heavier elements took form by neutron capture. Gamow and Alpher believed that all elements, up to uranium, were built up this way. Therefore, as Gamow wrote in 1948 [2] [3]:

Since the building-up process must have started with the formation of deuterons from the primordial neutrons and the protons into which some of these neutrons have decayed, we conclude that the temperature at that time must have been of the order $T_{nuc} \approx 10^9 K$ (which corresponds to the dissociation energy of deuterium nuclei).

Gamow also assumed that the Universe is 3 billion years old (too small by a factor of 4 by current standards), i.e. to $t_0 \approx 10$ Gyr, and the current density of matter is $\rho_{mat}(t_0) = 10^{-30} gcm^{-3}$ (too small by a factor of 8 from current estimates). These values, like the general concept of Friedmann's open Universe, had been generally accepted in the cosmology of the fifties.

Exercise 2

To determine how the uncertainty in the value of the baryon-to-photon ratio η affects the recombination temperature T_{rec} in the early Universe, we must find this temperature from the Saha equation. The Saha equation gives the fractional ionization X:

$$\frac{1-X}{X} = \frac{2\zeta(3)}{\pi^2}\eta \left(\frac{2\pi kT}{c^2 m_e}\right)^{\frac{3}{2}} e^{\frac{B_H}{kT}} \tag{11}$$

where T is the temperature, that we consider in the range 2000 K $<$ T $<$ 5500 K, B_H is the binding energy of hydrogen, that is 13.6 eV and η is the baryon-to-photon ratio. Hence, we plot the fractional ionization X as function of temperature and we obtain the solutions show in figure (1) for $\eta = 10^{-10}$ and $\eta = 10^{-9}$. Then, the recombination temperature T_{rec} is taken to be when $X = 0.1$, so for $\eta = 10^{-10}$ we have $T_{rec} = 3358.94K$ and for $\eta = 10^{-9}$ we have $T_{rec} = 3538.21K$. The change in η has a small effects on the temperature, that is a relative change of about 5%.

Exercise 3

Using the Euler equations we demonstrate that the vorticity of the peculiar velocity field in a theory of structure formation via gravitational collapse of small inhomogeneities decays away

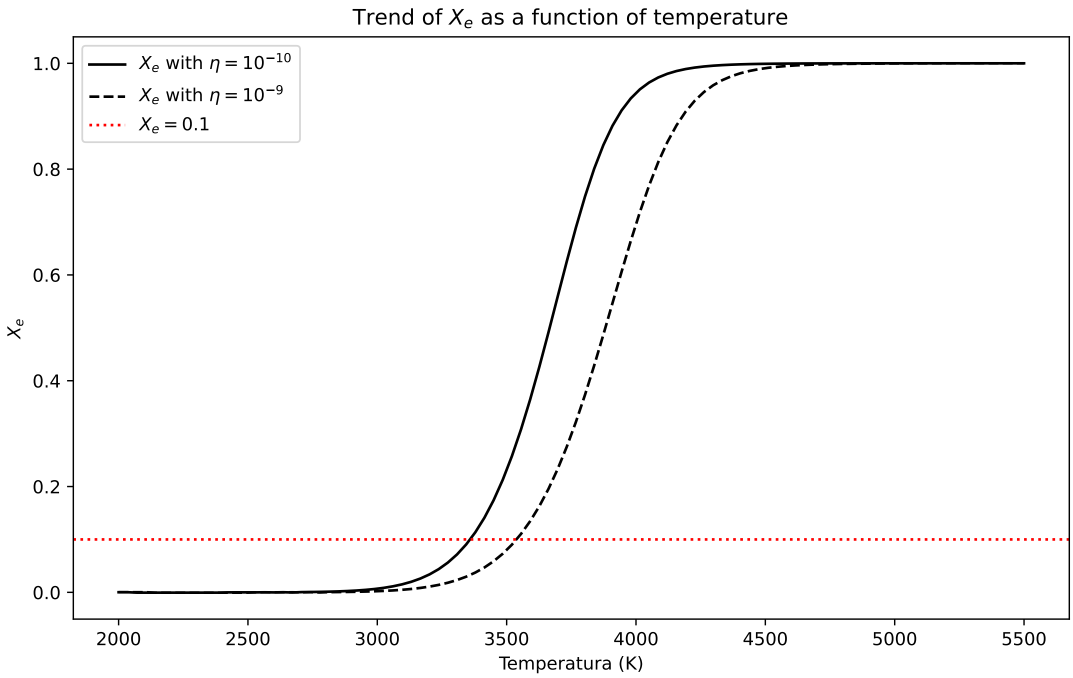

Figure 1: Plot of fractional ionization X as a function of temperature T (in Kelvin) for baryon-to-photon ratios $\eta = 10^{-10}$ and $\eta = 10^{-9}$. The code can be found in Appendix A.

like $1/a$. Mathematically, the vorticity $\vec{\omega}$ is the curl of the velocity field $\vec{u}$:

$$\vec{\omega} = \nabla \times \vec{u} \tag{12}$$

We start with the equation of conservation of momentum for non relativistic fluid, in comoving coordinates and with the approximation of small inhomogeneities:

$$\frac{\partial \vec{u}}{\partial t} + \frac{\dot{a}}{a}\vec{u} = -\frac{1}{a}\nabla_x \Phi - \frac{c_s^2}{a}\nabla_x \delta \tag{13}$$

where a is the scale factor, Φ is the peculiar potential, c_s^2 is the sound velocity and $\delta = \frac{\rho - \bar{\rho}}{\bar{\rho}}$ is the density contrast with ρ density perturbation and $\bar{\rho}$ the mean density of background. Now we consider that we are in a theory of formation of structures by gravitational collapse, this implies that the force of gravity dominates over the force of pressure. The gravitational force is given by the term containing the derivative of the gravitational potential Φ while the pressure force is given by the term that is proportional to the speed of sound c_s. Therefore, we can neglect the last term in the equation (13), that becomes:

$$\frac{\partial \vec{u}}{\partial t} + \frac{\dot{a}}{a}\vec{u} = -\frac{1}{a}\nabla_x \Phi \tag{14}$$

We applying the rotor to the right and left of the equation (14):

$$\frac{\partial \nabla \times \vec{u}}{\partial t} + \frac{\dot{a}}{a}\nabla \times \vec{u} = -\frac{1}{a}\nabla \times \nabla_x \Phi \tag{15}$$

so we can introduce the vorticity $\vec{\omega}$:

$$\frac{\partial \vec{\omega}}{\partial t} + \frac{\dot{a}}{a}\vec{\omega} = 0 \tag{16}$$

where we have used that $\nabla \times \nabla_x \Phi = 0$. At this point we have a differential equation to solve, so let's integrate:

$$\int \frac{1}{\vec{\omega}} d\vec{\omega} = -\int \frac{1}{a} da \tag{17}$$

which has solution:

$$\vec{\omega} = \vec{\omega}_0 \frac{a_0}{a} \tag{18}$$

Thus, we conclude that the vorticity decays away like $1/a$. Moreover, we show that in the absence of external forces the velocity field itself decays like $1/a$ too. In absence of external forces the equation of conservation of momentum (13) becomes:

$$\frac{\partial \vec{u}}{\partial t} + \frac{\dot{a}}{a}\vec{u} = 0 \tag{19}$$

Hence we have the same differential equation of (16), that has solution:

$$\vec{u} = \vec{u_0} \frac{a_0}{a} \tag{20}$$

Now we want to show that the peculiar velocity $\vec{u}$ is proportional to the acceleration induced by the density field. Let's start with the linearized mass conservation equation in comoving coordinates, that is:

$$\frac{\partial \delta(\vec{x}, t)}{\partial t} + \frac{1}{a}\nabla_x \cdot \vec{u} = 0 \tag{21}$$

where $\delta(\vec{x}, t)$ is the density contrast. The task is considerably streamlined by converting this equation into Fourier space, enabling the independent treatment of each wavelength. The Fourier transformation of the density contrast and the peculiar velocity yields, correspondingly:

$$\begin{aligned} \delta(\vec{x}, t) &= \sum_{\vec{k}} \delta_k(t) e^{-i\vec{k}\cdot\vec{x}} \\ \vec{u}(\vec{x}, t) &= \sum_{\vec{k}} \vec{u}_k(t) e^{-i\vec{k}\cdot\vec{x}} \end{aligned} \tag{22}$$

By inserting these two equations into equation (21), the resulting peculiar velocities are:

$$\frac{\partial \delta_k}{\partial t} = \frac{i}{a}\vec{k} \cdot \vec{u}_k \implies \vec{u}_k = -ia \frac{\partial \delta_k}{\partial t} \frac{\vec{k}}{k^2} \tag{23}$$

Now $\partial \delta_k / \partial t$ can be rewritten as:

$$\frac{\partial \delta_k}{\partial t} = \frac{\partial \delta_k}{\partial a}\frac{\partial a}{\partial t} = \frac{\partial \delta_k}{\partial a}\dot{a} = \frac{a}{a}\frac{\partial \delta_k}{\partial a}\dot{a}\frac{\delta_k}{\delta_k} = \frac{\dot{a}}{a}\delta_k f(a) \quad \text{with} \quad f(a) = \frac{\partial \ln \delta_k}{\partial \ln a} \tag{24}$$

where in the first passage we changed the partial derivative from t to a, then we multiplied and divided by $a\delta_k/a\delta_k$ and finally we have defined $f(a)$. By combining equation (24) and (23), one have an expression for the relation between velocity and density:

$$\vec{u}_k = -ia\delta_k f(a)\frac{\vec{k}}{k^2} \tag{25}$$

In the gravitational instability picture one assumes that it is gravity that makes the perturbations grow. It is therefore natural to introduce the gravitational potential instead of the density contrast δ_k. The linearized Poisson equation in comoving coordinates is:

$$\nabla^2\Phi = 4\pi G\bar{\rho}a^2\delta \tag{26}$$

where $\bar{\rho}$ is the background density. The Fourier transfomration of the peculiar potential is:

$$\Phi(\vec{x},t) = \sum_{\vec{k}} \Phi_k(t)e^{-i\vec{k}\cdot\vec{x}} \tag{27}$$

Putting (27) into (26) we obtain:

$$\Phi_k k^2 = -4\pi G\bar{\rho}a^2\delta_k \implies \delta_k = -\frac{k^2}{4\pi G\bar{\rho}a^2}\Phi_k \tag{28}$$

Inserting this into equation (25) results in an expression for the connection between the velocity field and the gravitational potential:

$$\vec{u}_k = \frac{i}{4\pi G\bar{\rho}a}\Phi_k f(a)\vec{k} = i\frac{2f(a)}{3H\Omega_m a}\Phi_k\vec{k} \tag{29}$$

where $\bar{\rho} = \frac{3H^2\Omega_m}{8\pi G}$ and since we are interested in (dark) matter perturbations, we can write Ω_m. Let us transfer it into real space:

$$\vec{u} = -\frac{2f(a)}{3H\Omega_m a}\nabla\Phi = \frac{2f(a)}{3H\Omega_m}\vec{g} \tag{30}$$

This equation show us that the velocity field is parallel and proportional to the peculiar acceleration, and then also to the gravitational force.

Exercise 4

The flat ΛCDM model is a mathematical model of the Big Bang theory with a cosmological constant denoted by lambda (Λ) associated with dark energy, the postulated cold dark matter, and the ordinary matter. Let's start with the evolution of density contrast for non relativistic fluid $\delta(\vec{x},t)$ in a homogeneous and isotropic background, in coomoving coordinates and for small perturbation is:

$$\partial_t^2\delta + 2H\partial_t\delta - 4\pi G\bar{\rho}\delta - \frac{c_s^2}{a^2}\Delta_x\delta = 0 \tag{31}$$

where the notation ∂_t is $\frac{\partial}{\partial t}$. We take a decomposition in plane waves as an ansatz for the solution as:

$$\delta(\vec{x},t) = \sum_{\vec{k}} \delta_k(t) e^{-i\vec{k}\cdot\vec{x}} \tag{32}$$

Doing this we are separating space and time. Putting (32) in (31), for dark matter ($c_s^2 = 0$), we obtain:

$$\partial_t^2 \delta(t) + 2H\partial_t \delta(t) - 4\pi G\bar{\rho}\delta(t) = 0 \tag{33}$$

where we decided to omit the subscript k to avoid making the notation heavier ($\delta_k(t) = \delta(t)$). Now we can rewrite (33) as:

$$\begin{aligned}
\partial_t^2 \delta(t) + 2H\partial_t \delta(t) - 4\pi G\Omega_m \rho_{crit}\delta(t) = 0 \\
\partial_t^2 \delta(t) + 2H\partial_t \delta(t) - 4\pi G\Omega_m \frac{3H^2}{8\pi G}\delta(t) = 0 \\
\partial_t^2 \delta(t) + 2H\partial_t \delta(t) - \frac{3}{2}\Omega_m H^2 \delta(t) = 0 \\
\partial_t(\partial_a(\delta)\dot{a}) + 2H\dot{a}\partial_a\delta(t) - \frac{3}{2}\Omega_m H^2 \delta(t) = 0 \\
\partial_a^2(\delta)\dot{a}^2 + \partial_a(\delta)\ddot{a} + 2H\dot{a}\partial_a\delta(t) - \frac{3}{2}\Omega_m H^2 \delta(t) = 0 \\
a^2\partial_a^2(\delta) + \left(\frac{\ddot{a}a^2}{\dot{a}^2} + 2a\right)\partial_a(\delta) - \frac{3}{2}\Omega_m \delta(t) = 0 \\
\partial_a^2(\delta) + \left(\frac{\ddot{a}}{\dot{a}^2} + \frac{2}{a}\right)\partial_a(\delta) - \frac{3}{2}\frac{\Omega_m}{a^2}\delta(t) = 0
\end{aligned} \tag{34}$$

From here, we have two possible ways forward: the first one is to rewrite the equation (34) as an hypergeometric equation. In fact, any ordinary second-order differential equation (with at most three regular singularities) can be transformed into the hypergeometric equation. On the other hand, we can start from the exact solution of equation (34), that is:

$$\delta(a) = \frac{5}{2}\Omega_{m,0}\frac{H}{H_0}\int_0^{\tilde{a}} \frac{1}{(aH)^3}da \tag{35}$$

and try to solve the integral. We do the second method here. The first can be found in Appendix B . Since we are in a flat ΛCDM model, we know the expression for H from the first Friedman equation:

$$H^2 = H_0^2(\Omega_{m,0}a^{-3} + (1-\Omega_{m,0})) \tag{36}$$

So (35) becomes:

$$\delta(a) = \frac{5}{2}\Omega_{m,0}\frac{H}{H_0}\int_0^{\tilde{a}} \frac{1}{a^3 H_0^3(\Omega_{m,0}a^{-3} + (1-\Omega_{m,0}))^{3/2}}da \tag{37}$$

Now we introduce the substitution:

$$u = \left(\frac{a}{\tilde{a}}\right)^3 \implies du = 3\frac{a^2}{(\tilde{a})^3}da \tag{38}$$

and (37) becomes:

$$
\begin{aligned}
\delta(a) &= \frac{5}{2}\Omega_{m,0}\frac{H}{H_0^4}\int_0^1 \frac{du\tilde{a}^3}{3u^{2/3}\tilde{a}^2}\frac{1}{u\tilde{a}^3\left(\frac{\Omega_{m,0}}{u\tilde{a}^3}+(1-\Omega_{m,0})\right)^{3/2}} \\
\delta(a) &= \frac{5}{2}\Omega_{m,0}\frac{H}{H_0^4}\int_0^1 \frac{du}{3u^{2/3}}\frac{1}{u\tilde{a}^2\left(\frac{\Omega_{m,0}}{u\tilde{a}^3}+(1-\Omega_{m,0})\right)^{3/2}} \\
\delta(a) &= \frac{5}{2}\Omega_{m,0}\frac{H}{H_0^4}\int_0^1 \frac{du}{3u^{5/3}\tilde{a}^2\left(\frac{\Omega_{m,0}}{u\tilde{a}^3}+(1-\Omega_{m,0})\right)^{3/2}} \\
\delta(a) &= \frac{5}{2}\Omega_{m,0}\frac{H}{H_0^4}\int_0^1 \frac{du}{3u^{5/3}\tilde{a}^2\frac{\Omega_{m,0}}{u^{3/2}\tilde{a}^{9/2}}\left(1+\frac{(1-\Omega_{m,0})}{\Omega_{m,0}}u\tilde{a}^3\right)^{3/2}} \\
\delta(a) &= \frac{5}{2}\Omega_{m,0}^{-\frac{1}{2}}\frac{H}{H_0^4}\int_0^1 \frac{du}{3u^{1/6}\tilde{a}^{-\frac{5}{2}}\left(1+\frac{(1-\Omega_{m,0})}{\Omega_{m,0}}u\tilde{a}^3\right)^{3/2}}
\end{aligned}
\tag{39}
$$

And the integral can be performed to give:

$$
\int_0^1 \frac{du}{u^{1/6}(1+xu)^{3/2}} = \frac{6}{5}u^{5/6}\ {}_2F_1\left(\frac{3}{2},\frac{5}{6};\frac{11}{6};-ux\right) \tag{40}
$$

with $x = \frac{(1-\Omega_{m,0})}{\Omega_{m,0}}\tilde{a}^3$. This integral can be recognised as an Euler-type representation of a hypergeometric function [4]:

$$
B(b,c-b)_2F_1(a,b;c;z) = \int_0^1 x^{b-1}(1-x)^{c-b-1}(1-zx)^{-a}dx \tag{41}
$$

where B is the beta function. Putting the integral (40) into equation (39) we obtain:

$$
\delta(a) = \frac{5}{2}\Omega_{m,0}^{-\frac{1}{2}}\frac{H}{H_0^4}\frac{1}{\tilde{a}^{-\frac{5}{2}}}\frac{2}{5}u^{5/6}\ {}_2F_1\left(\frac{3}{2},\frac{5}{6};\frac{11}{6};-ux\right) \tag{42}
$$

Now we simplify the prefactor before the hypergeometric function:

$$
\begin{aligned}
\delta(a) = \Omega_{m,0}^{-\frac{1}{2}}\frac{H_0(\Omega_{m,0}a^{-3}+(1-\Omega_{m,0}))^{1/2}}{H_0^4}\frac{1}{\tilde{a}^{-\frac{5}{2}}}\left(\frac{a}{\tilde{a}}\right)^{5/2}\ {}_2F_1\left(\frac{3}{2},\frac{5}{6};\frac{11}{6};-a^3\frac{(1-\Omega_{m,0})}{\Omega_{m,0}}\right) \\
\delta(a) = \frac{(1-a^3(1-\Omega_{m,0}^{-1}))^{1/2}}{H_0^3}a^{-\frac{3}{2}}a^{\frac{5}{2}}\ {}_2F_1\left(\frac{3}{2},\frac{5}{6};\frac{11}{6};a^3(1-\Omega_{m,0}^{-1})\right)
\end{aligned}
\tag{43}
$$

In an equivalent form the hypergeometric function can be written as [4]:

$$
{}_2F_1(a,b;c;z) = (1-z)^{c-a-b}\ {}_2F_1(c-a,c-b;c;z) \tag{44}
$$

Hence, the growth factor is:

$$\delta(a) = (1 - a^3(1 - \Omega_{m,0}^{-1}))^{-1/2} \frac{(1 - a^3(1 - \Omega_{m,0}^{-1}))^{1/2}}{H_0^3} a \; {}_2F_1\left(\frac{1}{3}, 1; \frac{11}{6}; a^3(1 - \Omega_{m,0}^{-1})\right)$$
$$\boxed{\delta(a) = \frac{a}{H_0^3} \; {}_2F_1\left(\frac{1}{3}, 1; \frac{11}{6}; a^3(1 - \Omega_{m,0}^{-1})\right)} \tag{45}$$

The luminosity distance is defined as:

$$d_L = c(1+z)\int_{t_1}^{t_0} \frac{dt}{a(t)} = c(1+z)\int_0^{\tilde{z}} \frac{dz}{H(z)} \tag{46}$$

In the flat ΛCDM cosmology the first Friedman equation is:

$$H^2(z) = H_0^2(\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}) \tag{47}$$

Thus, the general expression for the luminosity distance is given by:

$$d_L(z) = \frac{c(1+z)}{H_0}\int_0^{\tilde{z}} \frac{dz}{\sqrt{\Omega_{m,0}(1+z)^3 + ((1 - \Omega_{m,0}))}} \tag{48}$$

where we have used that $\Omega_{\Lambda,0} = 1 - \Omega_{m,0}$ and $\Omega_{m,0}$ is the current matter density of the Universe. If we introduce:

$$s = \sqrt[3]{\frac{(1 - \Omega_{m,0})}{\Omega_{m,0}}} \tag{49}$$

and we apply the substitution:

$$u = \frac{s}{1+z} \implies du = -\frac{s}{(1+z)^2}dz = -\frac{u^2}{s}dz \tag{50}$$

the equation (48) can conveniently be written as:

$$d_L = -\frac{c(1+z)}{H_0}\int_s^{\frac{s}{1+z}} \frac{s}{u^2\sqrt{\Omega_{m,0}\left(\frac{s^3}{u^3} + s^3\right)}}du$$
$$d_L = -\frac{c(1+z)}{\sqrt{\Omega_{m,0}s}H_0}\int_s^{\frac{s}{1+z}} \frac{\sqrt{u^3}}{u^2\sqrt{1+u^3}}du \tag{51}$$
$$d_L = -\frac{c(1+z)}{\sqrt{\Omega_{m,0}s}H_0}\int_s^{\frac{s}{1+z}} \frac{1}{\sqrt{u(1+u^3)}}du$$

Now we divide the integral in two parts:

$$d_L = -\frac{c(1+z)}{\sqrt{\Omega_{m,0}s}H_0}\left[\int_s^0 \frac{1}{\sqrt{u(1+u^3)}}du + \int_0^{\frac{s}{1+z}} \frac{1}{\sqrt{u(1+u^3)}}du\right]$$
$$d_L = \frac{c(1+z)}{\sqrt{\Omega_{m,0}s}H_0}\left[\int_0^s \frac{1}{\sqrt{u(1+u^3)}}du - \int_0^{\frac{s}{1+z}} \frac{1}{\sqrt{u(1+u^3)}}du\right] \tag{52}$$
$$d_L = \frac{c(1+z)}{\sqrt{\Omega_{m,0}s}H_0}\left[A(s) - A\left(\frac{s}{1+z}\right)\right]$$

with the function $A(x)$ defined as:

$$A(x) = \int_0^x \frac{1}{\sqrt{u(1+u^3)}} du \tag{53}$$

Now, to obtain the expressions (53) in terms of hypergeometric functions we apply the transformation:

$$t = \left(\frac{u}{x}\right)^3 \implies dt = 3\frac{u^2}{x^3} du \tag{54}$$

This yields:

$$\begin{aligned} A(x) &= \int_0^1 \frac{dt}{3u^2} x^3 \frac{1}{\sqrt{t^{1/3} x(1+tx^3)}} \\ A(x) &= \int_0^1 \frac{dt}{3x^2 t^{2/3}} x^{\frac{5}{2}} t^{-\frac{1}{6}} \frac{1}{\sqrt{1+tx^3}} \\ A(x) &= \int_0^1 \frac{x^{1/2}}{3t^{5/6}(1+x^3 t)^{1/2}} dt = \frac{1}{3}\sqrt{x} \int_0^1 \frac{1}{t^{5/6}(1+x^3 t)^{1/2}} dt \end{aligned} \tag{55}$$

This integral can be recognised as an Euler-type representation of a hypergeometric function, as in equation (41). Therefore, the results is the simple expression:

$$A(x) = 2\sqrt{x}\,{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; -x^3\right) \tag{56}$$

Now we come back to the luminosity distance:

$$\begin{aligned} d_L &= \frac{c(1+z)}{\sqrt{\Omega_{m,0} s} H_0} \left[A(s) - A\left(\frac{s}{1+z}\right)\right] \\ d_L &= \frac{c}{a\sqrt{\Omega_{m,0} s} H_0} \left[2\sqrt{s}\,{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; -s^3\right) - 2\sqrt{as}\,{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; -(as)^3\right)\right] \\ d_L &= \frac{2c}{a\sqrt{\Omega_{m,0}} H_0} \left[{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; -s^3\right) - \sqrt{a}\,{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; -(as)^3\right)\right] \end{aligned} \tag{57}$$

Now we recall that the parameter s has the form (49), so we find the luminosity distance:

$$\boxed{d_L = \frac{2c}{a\sqrt{\Omega_{m,0}} H_0} \left[{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; 1 - \Omega_{m,0}^{-1}\right) - \sqrt{a}\,{}_2F_1\left(\frac{1}{6}, \frac{1}{2}; \frac{7}{6}; a^3(1 - \Omega_{m,0}^{-1})\right)\right]} \tag{58}$$

Appendix A

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import root_scalar

# Funzione di Saha semplificata
def saha_equation_simplified(Xe, T, eta):
    Bh = 13.6 * 1.602176634e-19  # Energia di legame dell'idrogeno [J]
    k = 1.380649e-23  # Costante di Boltzmann [J/K]
    c = 299792458  # Velocità della luce nel vuoto [m/s]
    me = 9.10938356e-31  # Massa dell'elettrone [kg]
    pi = np.pi

    exp_term = np.exp(Bh / (k * T))
    sqrt_term = np.sqrt((2 * np.pi * k * T) / ( c**2 * me))
    return (1 - Xe) / Xe**2 - 2*2.6*(eta)/pi**2 * sqrt_term**3 * exp_term

# Funzione per trovare la temperatura corrispondente a Xe = 0.1
def find_temperature(Xe_target, eta):
    def equation_to_solve(T):
        return saha_equation_simplified(Xe_target, T, eta)

    sol = root_scalar(equation_to_solve, bracket=[100, 20000])
    return sol.root

# Range di temperature
temperatures = np.linspace(2000, 5500, 100)  # da 100 a 20000 K

# Calcoliamo Xe per ogni temperatura per eta = 10^-10
Xe_values_eta_1e10 = []
for T in temperatures:
    sol = root(equation_to_solve, 0.001, args=(T, 1e-10))
    Xe_values_eta_1e10.append(sol.x[0])

# Calcoliamo Xe per ogni temperatura per eta = 10^-9
Xe_values_eta_1e9 = []
for T in temperatures:
    sol = root(equation_to_solve, 0.001, args=(T, 1e-9))
    Xe_values_eta_1e9.append(sol.x[0])

# Troviamo la temperatura corrispondente a Xe = 0.1 per entrambi i valori di eta
temperature_eta_1e10 = find_temperature(0.1, 1e-10)
```

```
temperature_eta_1e9 = find_temperature(0.1, 1e-9)

# Plot
plt.figure(figsize=(10, 6))
plt.plot(temperatures, Xe_values_eta_1e10, label='$X_e$ with $\\eta = 10^{-10}$',
linestyle='-', color='black')
plt.plot(temperatures, Xe_values_eta_1e9, label='$X_e$ with $\\eta = 10^{-9}$',
linestyle='--', color='black')
plt.axhline(y=0.1, color='red', linestyle=':', label='$X_e = 0.1$')
plt.xlabel('Temperatura (K)')
plt.ylabel('$X_e$')
plt.title('Trend of $X_e$ as a function of temperature')
plt.legend()
plt.savefig('Saha.pdf')
plt.show()

print(f"Temperatura corrispondente a Xe = 0.1 per eta = 10^-10:
{temperature_eta_1e10:.2f} K")
print(f"Temperatura corrispondente a Xe = 0.1 per eta = 10^-9:
{temperature_eta_1e9:.2f} K")
```

Appendix B

From equation (34), we recall that:

$$H = \frac{\dot{a}}{a} \implies \dot{H} = \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} \tag{59}$$

and continuing from where we stopped in (34), we obtain:

$$\begin{aligned}
\partial_a^2(\delta) + \left(\frac{\ddot{a}}{\dot{a}^2}\cdot\frac{a}{a} + \frac{2}{a}\right)\partial_a(\delta) - \frac{3}{2}\frac{\Omega_m}{a^2}\delta(t) = 0 \\
\partial_a^2(\delta) + \left[\frac{a}{\dot{a}^2}\left(\dot{H} + \frac{\dot{a}^2}{a^2}\right) + \frac{2}{a}\right]\partial_a(\delta) - \frac{3}{2}\frac{\Omega_m}{a^2}\delta(t) = 0 \\
\partial_a^2(\delta) + \left(\frac{\dot{H}a}{\dot{a}^2} + \frac{3}{a}\right)\partial_a(\delta) - \frac{3}{2}\frac{\Omega_m}{a^2}\delta(t) = 0 \\
\partial_a^2(\delta) + \left(\frac{\dot{H}}{H\dot{a}} + \frac{3}{a}\right)\partial_a(\delta) - \frac{3}{2}\frac{\Omega_m}{a^2}\delta(t) = 0 \\
\frac{d^2\delta}{da^2} + \left(\frac{d\ln H}{da} + \frac{3}{a}\right)\frac{d\delta}{da} - \frac{3}{2}\frac{\Omega_m}{a^2}\delta(t) = 0
\end{aligned} \tag{60}$$

where in the last passage, we have changed from partial derivative to total derivative since δ depends only on t i.e. $\delta(t)$ (or in this case on a, i.e. $\delta(a)$). This change could have been made as early as formula (33). We are able to find the exact analytic growing mode solution of δ, for a constant dark energy equation of state parameter, ω as follows [5] [6]. By noting that the density contrast for pure matter grows like $\delta(a) \propto a$, one can write the general function:

$$\delta(a) = aG(a) \implies \frac{d\delta}{da} = G(a) + aG'(a) \implies \frac{d^2\delta}{da^2} = 2G'(a) + aG''(a) \tag{61}$$

where $'$ means $\frac{d}{da}$, so with which definition the equation (60) becomes:

$$a^2G''(a) + \left(5a + \frac{a^2H'}{H}\right)G'(a) + \left(3 + a\frac{H'}{H} - \frac{3}{2}\Omega_m\right)G(a) = 0 \tag{62}$$

Now we introduce one new parameter in equation (62):

$$u = -\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}a^{-3\omega} \implies a\frac{d}{da} = -3\omega u\frac{d}{du} \implies \Omega_m(a) = \frac{1}{1-u} \tag{63}$$

and we get:

$$u(1-u)G''(u) + \left[1 - \frac{5}{6\omega} - \left(\frac{3}{2} - \frac{5}{6\omega}\right)u\right]G'(u) - \frac{1-\omega}{6\omega^2}G(u) = 0 \tag{64}$$

The above equation becomes the so called hypergeometric equation with the complete solution:

$$\begin{aligned}
\delta(a) = C_1 a\ {}_2F_1\left(\frac{\omega-1}{2\omega}, -\frac{1}{3\omega}, 1 - \frac{5}{6\omega}; -\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}a^{-3\omega}\right) \\
+C_2 a^{-3/2}\ {}_2F_1\left(\frac{1}{2\omega}, \frac{1}{2} + \frac{1}{3\omega}, 1 + \frac{5}{6\omega}; -\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}a^{-3\omega}\right)
\end{aligned} \tag{65}$$

The initial term represents the growing mode solution, while the subsequent term represents the decaying mode. In discussing solutions at late times, we consistently consider the growing mode solution.

References

[1] S. Dodelson, *Modern Cosmology*. San Diego: Academic Press, 2003.

[2] R. A. Alpher, H. Bethe, and G. Gamow, "The origin of chemical elements," *Physical Review*, vol. 73, no. 7, p. 803, 1948.

[3] P. J. E. Peebles, "Discovery of the hot big bang: What happened in 1948," *The European Physical Journal H*, vol. 39, no. 2, pp. 205–223, 2014.

[4] M. Abramowitz, *Handbook of Mathematical Functions, With Formulas, Graphs, and Mathematical Tables,*. USA: Dover Publications, Inc., 1974.

[5] A. B. Belloso, J. Garcia-Bellido, and D. Sapone, "A parametrization of the growth index of matter perturbations in various dark energy models and observational prospects using a euclid-like survey," *Journal of Cosmology and Astroparticle Physics*, vol. 2011, no. 10, p. 010, 2011.

[6] S. Lee and K.-W. Ng, "Growth index with the exact analytic solution of sub-horizon scale linear perturbation for dark energy models with constant equation of state," *Physics Letters B*, vol. 688, no. 1, pp. 1–3, 2010.